Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Classical uncertainty principle

Files.

- File: `briefs/ClassicalUncertaintyPrinciple/ClassicalUncertaintyPrinciple.lean`
- Library: `ClassicalUncertaintyPrinciple`
- Imports: two new `ForMathlib` files:
  `ForMathlib/InformationTheory/GibbsInequality.lean`: Gibbs' inequality $-\int \rho \log \rho \leq -\int \rho \log G$ for a probability density ρ and a positive G with $\int G \leq 1$ (`MeasureTheory.integral_negMulLog_le`), with equality only for $\rho = G$ a.e. (`MeasureTheory.ae_eq_of_integral_negMulLog_eq`);
  `ForMathlib/Probability/Distributions/Gaussian/Real.lean`: the logarithm of the Gaussian density (`log_gaussianPDFReal`), and its mean and variance as Lebesgue integrals (`integral_mul_gaussianPDFReal`, `integral_sub_sq_mul_gaussianPDFReal`).

Setting. An ensemble on a measure space (X, μ) (`IsEnsemble` ρ μ) is a function $\rho \geq 0$ with $\int \rho \, d\mu = 1$ and $\rho \log \rho$ integrable (finite entropy). Its entropy relative to a cell of size c is `entropy` c ρ $\mu = -\int \rho \log(c\rho) \, d\mu$, in nats. A variable $f : X \to \mathbb{R}$ has mean `mean`, variance `var` and standard deviation `sd` under ρ. One DOF has phase space $\mathbb{R} \times \mathbb{R}$ with Lebesgue measure and cell $h > 0$, and $\hbar = h/2\pi$ (`hbar`). The conditions are:

- `LowerEntropicBound` h ρ (ZENT): $S_h[\rho] \geq 0$;
- `UncertaintyPrinciple` h ρ (CUP): $\sigma_q \sigma_p \geq \hbar/e$.

Finite second moments of the variables are hypotheses of the theorems. The Poisson bracket `poissonBracket` f g is $\partial_q f \, \partial_p g - \partial_p f \, \partial_q g$, with Fréchet derivatives.

Results.

Brief	Lean
Corollary 1, $S[\rho_U] = \log(\mu(U)/h^n)$	`entropy_uniform` (any measure space, any cell $c > 0$); `entropy_uniform_phaseSpace` (phase space $(\mathbb{R}^2)^n$, $c = h^n$)
Corollary 1, $S[\rho_U] = 0$ iff $\mu(U) = h^n$	`entropy_uniform_eq_zero_iff`
ZENT, CUP	`LowerEntropicBound`, `UncertaintyPrinciple`
Theorem 2, proof: a product of two independent Gaussians minimizes $\sigma_q \sigma_p$ at given entropy, and then $\sigma_q \sigma_p = (\hbar/e)e^S$	`hbar_div_exp_mul_exp_entropy_le`: $\sigma_q \sigma_p \geq (\hbar/e)e^{S_h[\rho]}$ for every ensemble; `sd_mul_sd_eq_iff`: equality iff ρ is a.e. a product of Gaussians
Theorem 2, eq. (3), ZENT implies CUP	`uncertaintyPrinciple_of_lowerEntropicBound`
CUP, equality case	`sd_mul_sd_eq_hbar_div_exp_iff`, corrected (see "Problems found")
Eq. (4), the Poisson bracket is the Jacobian determinant	`det_prod`; `poissonBracket_fst_snd`: $\{q, p\} = 1$
Theorem 2, eq. (5), $\sigma_f \sigma_g \geq (\hbar/e)e^{E[\log \lvert\{f,g\}\rvert]}$ under ZENT	`hbar_div_exp_mul_exp_le_sd_mul_sd`; for every ensemble $\sigma_f \sigma_g \geq (\hbar/e)e^{S_h[\rho]+E[\log \lvert\{f,g\}\rvert]}$: `hbar_div_exp_mul_exp_entropy_add_le`
Remark, CUP does not imply ZENT	`exists_not_lowerEntropicBound`
(used for both equations) the Gaussian bound on any measure space	`Hyp.hbar_div_exp_mul_exp_le`: $\sigma_f \sigma_g \geq (\hbar/e)e^{S_h[\rho]+E[\log J]}$ for a weight $J > 0$ that turns every product of Gaussians in (f, g) into a subprobability density (`IsGaussWeight`)

Differences from the brief.

- The Lagrange-multiplier argument is replaced by Gibbs' inequality against the product G of Gaussians in q and p with the means and variances of ρ: $-\int \rho \log \rho \leq -\int \rho \log G = \log(2\pi e \, \sigma_q \sigma_p)$. This proves that the Gaussian is a minimum, not only a critical point, and gives the equality case. Only the marginal variances enter, so no covariance matrix is needed.
- Only one DOF is formalized: ZENT and CUP are stated for a density on $\mathbb{R}^2$ (the marginal of the DOF). Marginals of a density on $(\mathbb{R}^2)^n$ are not formalized; Corollary 1 is, for any n.

- The hypotheses that the brief leaves implicit are explicit: $\rho \geq 0$, $\int \rho = 1$, finite entropy and finite second moments. Positive variances are proved, not assumed: an ensemble of finite entropy cannot have zero variance in any variable, since Gibbs' inequality against ever narrower Gaussians would bound its entropy by $-\infty$.

- For eq. (5) the brief passes to the density ρ_{fg} of (f, g), with $\rho_{qp} = \rho_{fg} \, |\{f, g\}|$. Here ρ_{fg} is not formed. The Gaussian in (f, g) is pulled back to phase space with the weight $|\{f, g\}|$, and the change-of-variables formula shows that its integral is at most 1. This needs f, g differentiable, $(q, p) \mapsto (f, g)$ injective (not surjective, not orientation preserving), $\{f, g\} \neq 0$ everywhere, and $\rho \log |\{f, g\}|$ integrable. σ_f is the standard deviation of f under ρ. Physical units are not modelled, so the brief's assumption on them is not needed.

- The counterexample for the remark is a uniform distribution on two squares $[L, L + \varepsilon]^2$ and $[-L - \varepsilon, -L]^2$, with $\varepsilon = \sqrt{h}/2$ and $L = \sqrt{h}$. Its entropy is $\log \frac{1}{2} < 0$ and $\sigma_q \sigma_p \geq h/2 \geq \hbar/e$. The brief instead correlates q and p.

Problems found in the brief. The main claim (ZENT implies CUP, eqs. (3) and (5)) is correct; the equality case is misstated.

- CUP says that equality holds "if and only if position and momentum are independent gaussian distributions", and the proof ends the same way. Independent Gaussians satisfy $\sigma_q \sigma_p = (\hbar/e) e^S$, which equals $\hbar/e$ only when $S = 0$. Under ZENT, $\sigma_q \sigma_p = \hbar/e$ holds exactly for products of Gaussians with $\sigma_q \sigma_p = \hbar/e$ (`sd_mul_sd_eq_hbar_div_exp_iff`).

- The Lagrange multipliers give only a stationary point of $\sigma_q^2 \sigma_p^2$ at fixed entropy. They show neither that it is a minimum nor that a minimizer exists, which the proof then uses ("the gaussian is the distribution that minimizes the uncertainty").

- In the derivation of eq. (5), "order preserving" should be "orientation preserving"; the last two lines of the chain of equalities are identical; and CUP writes $E[\ln\{f, g\}]$ where the abstract and eq. (5) have $E[\ln |\{f, g\}|]$. The logarithm needs the absolute value when $\{f, g\} < 0$.

- The brief does not state that the entropy and the second moments must be finite. Without this $S[\rho]$ or σ may be undefined.